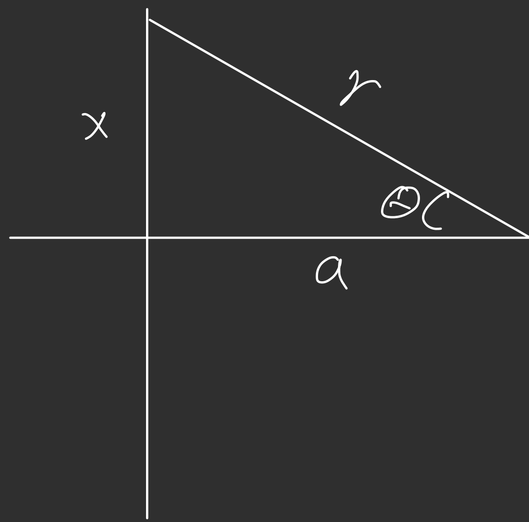
$$dE = \frac{k\,dq}{r^2}$$

$$dE_y = \frac{k\,dq}{r^2} \cdot \cos\theta$$

$$dE_x = \frac{k\,dq}{r^2} \cdot \sin\theta$$




$$r = \sqrt{x^2 + a^2}$$

$$dq = \lambda\,dx$$

$$dE_y = \frac{k\lambda\,dx}{(x^2+a^2)} \cdot \cos\theta$$

$$dE_y = \frac{k\lambda\,dx}{(x^2+a^2)} \cdot \frac{a}{\sqrt{(x^2+a^2)}}$$

$$dE_y = \frac{k\lambda a\,dx}{(x^2+a^2)^{3/2}}$$

$$E_y = \int_{-L}^{L} \frac{k\lambda a\,dx}{(x^2+a^2)^{3/2}} \qquad E_x = 0$$

$$x = a\tan\theta \qquad \theta = \tan^{-1}\frac{x}{a}$$

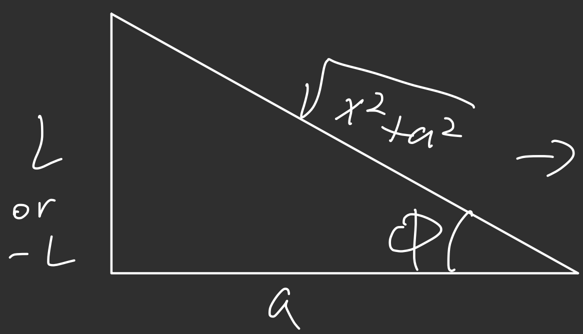
$$dx = a\sec^2\theta \cdot d\theta \qquad (x^2+a^2)^{\frac{3}{2}} = a^3\sec^3\theta$$

$$E_y = \int_{\tan^{-1}\frac{-L}{a}}^{\tan^{-1}\frac{L}{a}} \frac{k\lambda a^2\sec^2\theta}{a^3 \cdot \sec^3\theta} \cdot d\theta$$

$$E_y = \frac{k\lambda}{a} \int_{\tan^{-1}\frac{-L}{a}}^{\tan^{-1}\frac{L}{a}} \cos\theta \cdot d\theta$$

$$= \frac{k\lambda}{a} (\sin\theta) \Big|_{\tan^{-1}\frac{-L}{a}}^{\tan^{-1}\frac{L}{a}}$$

$$= \frac{k\lambda}{a} \left( \sin\left(\tan^{-1}\frac{L}{a}\right) - \sin\left(\tan^{-1}\frac{-L}{a}\right) \right)$$




$$\rightarrow \sin\phi = \frac{L}{\sqrt{x^2+a^2}}$$

or

$$= \frac{-L}{\sqrt{x^2+a^2}}$$

$$= \frac{2k\lambda L}{a\sqrt{L^2+a^2}}$$

as $L$ approaches $\infty$

$$= \frac{2k\lambda}{a}$$